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Smartphone Based Liquid Level Monitoring System

Measurement of liquid level in process tanks for monitoring and/or controlling is essential in process industries. In process plants, tanks contain various liquids that are generally costly. Some of the liquids are inflammable and corrosive as well. Therefore, it is very much essential to monitor the level of liquid in a tank so that it doesn't overflow from the tank.

Various methods are used to monitor the level of liquids, and the selection of technology to be used is based on several factors like:

- Maximum and minimum temperature
- Maximum and minimum pressure
- Tank geometry
- Nature of liquid



Fig. 1: Authors' prototype of the project

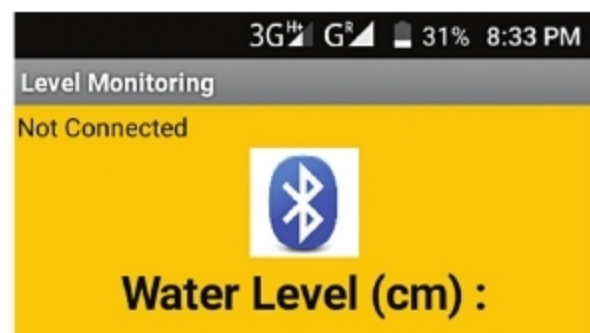


Fig. 2: Screenshot of the Android application

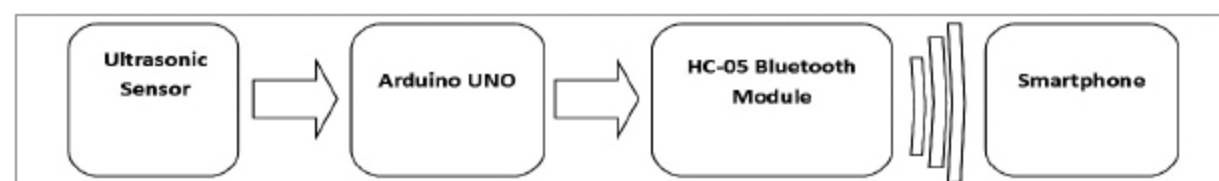


Fig. 3: Block diagram of the project

There are no level detectors or switches that can precisely specify the accuracy or reliability of a particular application. One of the most popular level detectors is the ultrasonic level detector. The advantages of ultrasonic level detector include its non-contact use and suitability for various liquids. Ultrasonic level detectors operate by generating an ultrasonic pulse and measuring the time taken for the echo to return.

If the detector is mounted at the top of the tank, the pulse travels in air at a speed of 331m/sec at 0°C. Therefore the time of travel is an indication of the depth of empty space above the liquid in the tank. In case of water at 25°C, an ultrasonic pulse travels at 1505m/sec. For testing, we mounted the detector on top of a glass jar and the liquid used was water. The author's prototype is shown in Fig. 1.

The screenshot of the Android application to monitor the water level in the tank is shown in Fig. 2. The block diagram of the project is shown in Fig. 3.

Circuit and working

The project consists of two parts: A transmitter unit built around Arduino Uno, an HC-05 Bluetooth module, and an ultrasonic sensor module (HC-SR04). The receiver unit is your Android smartphone with an Android app installed. The circuit diagram of the project (transmitter) is shown in Fig. 5. The major components used in this project are:

Android application. The level monitoring application (levelmonitoring.apk) used here has been built using the MIT App Inventor. When

this app is running, the data sent by the HC-05 Bluetooth module is received and displayed on the Android smartphone. The level monitoring application has been tested on Micromax Q372 smartphone with Android Lollipop v5.0.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller based development board with six analogue input pins and fourteen digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI, and I2C. The microcontroller can operate at a clock frequency of 16MHz. The HC-05 Bluetooth module and ultrasonic sensor module (HC-SR04) are interfaced with the Arduino Uno.

HC-05 Bluetooth module. It is an easy to use Bluetooth SPP (serial port protocol) module designed for transparent wireless serial connection setup. The serial port Bluetooth module is a fully qualified V2.0 Bluetooth 3Mbps EDR (enhanced data rate) modulation with a complete 2.4GHz radio transceiver and baseband. It uses the CSR Bluecore 04-External single-chip Bluetooth system with CMOS technology and AFH (adaptive frequency hopping feature). The top view of a typical 6-pin HC-05 Bluetooth module is shown in Fig. 4. This project uses only four pins: RXD, TXD, Vcc, and GND pins of the HC-05 module. So, you can use any similar 4-pin Bluetooth module for this project.

HC-SR04 ultrasonic sensor module. The ultrasonic ranging module HC-SR04 provides a 2cm to 400cm non-contact measurement func-

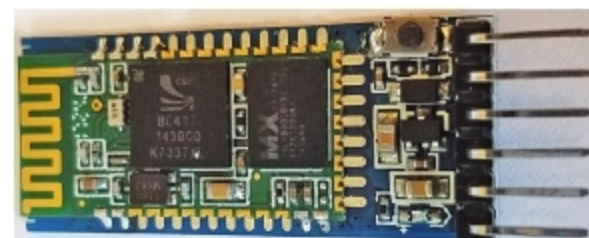


Fig. 4: Top view of typical HC-05 Bluetooth module